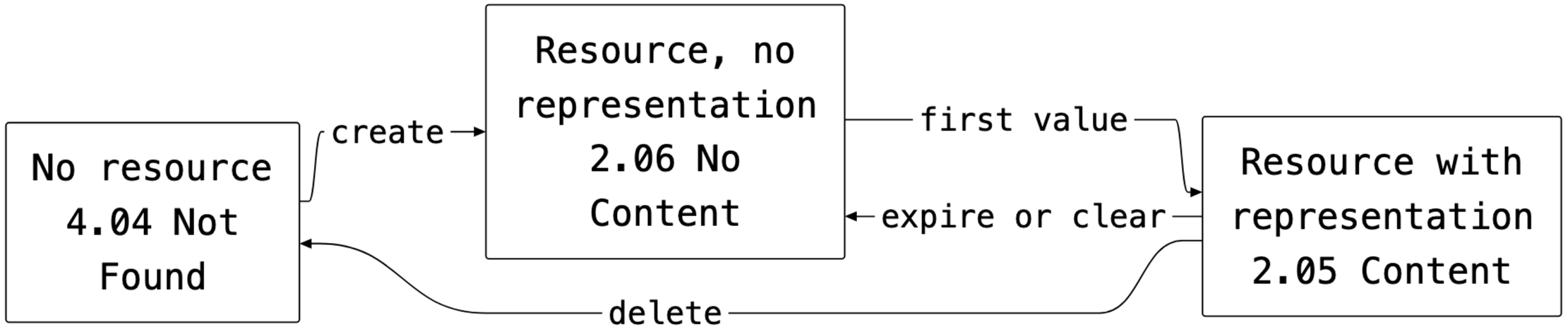
2.06 No Content

draft-jimenez-core-coap-no-content

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CoRE WG, October 2026

Resource states



Motivation

- In draft-ietf-core-coap-pubsub, a topic-data resource exists before the first publication
- The pubsub draft works around this with a half-created topic state, 4.04 responses, and an initialize property for placeholder data
- A sensor whose last reading has expired, or a value that has been cleared, is in the same state

Current practice

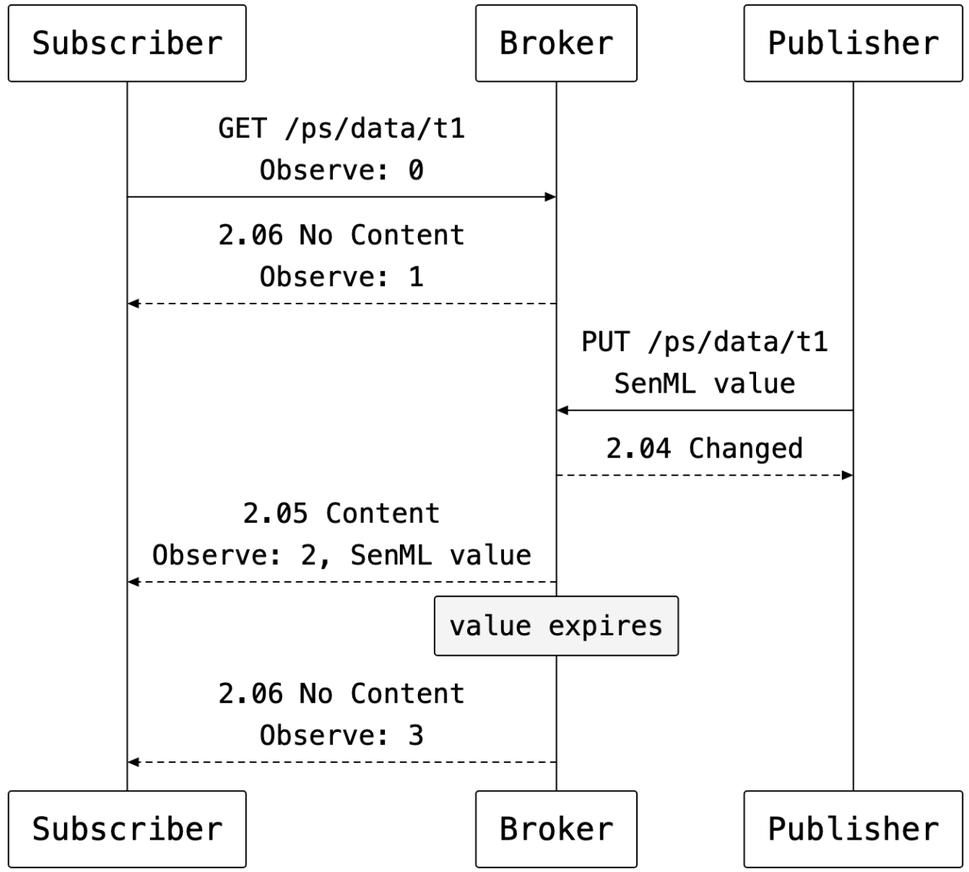
Response	Consequence
4.04 Not Found	Terminates the Observe registration; the client polls
2.05 Content, empty payload	Ambiguous: an empty payload is a valid representation in some formats
5.03 Service Unavailable	Reports a server failure that has not occurred

Semantics

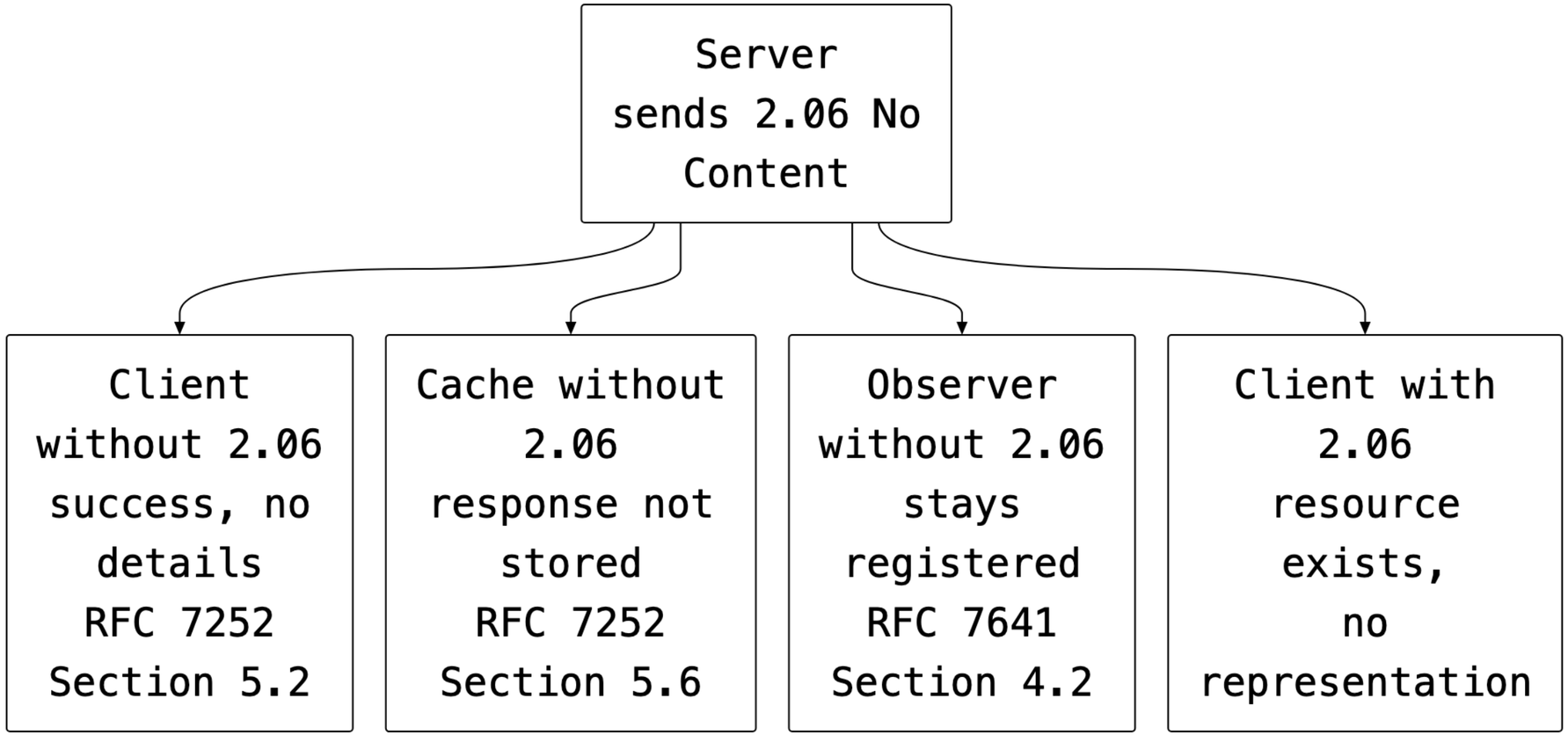
- The request succeeded, the target resource exists, and it has no current representation
- No payload, no Content-Format Option, no ETag Option
- Used with GET and FETCH; PUT, POST, PATCH, iPATCH and DELETE keep 2.04 and 2.02
- Cacheable; Max-Age indicates when a representation may become available
- A client holding an earlier representation no longer treats it as current

Observe

- A registration on a resource without a representation is confirmed with 2.06 and an Observe Option
- When a representation appears, a 2.05 notification follows
- When it disappears, a 2.06 notification keeps the registration and invalidates the previous value
- A client that requires a Content-Format includes Accept in the registration; the server rejects with 4.06



Endpoints without 2.06



Proxies and HTTP mapping

- 2.06 is end-to-end; a proxy does not translate it per client
- A proxy without 2.06 forwards the response and does not cache it
- HTTP-to-CoAP cross-proxy (RFC 8075): 2.06 ➳ 204 No Content
- CoAP-to-HTTP cross-proxy: 204 on GET ➳ 2.06; RFC 7252 Section 10.1.1 currently yields an empty 2.05

Design decisions

- Servers send 2.06 unconditionally; endpoints without it see the equivalent of an empty 2.05
- 2.06 means no representation in any format; a representation in a format the client did not request is a matter for Accept and 4.06
- RFC 7252 and RFC 7641 are the normative references; FETCH, the HTTP mapping and RFC 9110 are informative
- Appendix A records the rejected alternatives: empty 2.05, 4.04, 5.03, 2.04 on GET, a placeholder value, a null media type, an RFC 8710 multipart wrapper, and a request option signalling support

Open issues

Issue	Question
#1	Does this document update RFC 7641?
#2	May a 2.06 carry an ETag and be validatable?
#3	No representation at all, or none of the intended type?
#4	Caches that store an unrecognized 2.xx

Next steps

- Resolve the open issues
- Submit -00 to the datatracker
- Request feedback on the CoRE mailing list

Q&A